AGENTIC AI GRAND FINALE PROBLEM STATEMENT

NetCoop AI

Multi-Agent Network Fault Diagnosis and Interoperability

DOMAIN	FORMAT
Artificial Intelligence • Computer Networks	24-hour hackathon

1. Background and Problem Statement

IIT Bhubaneswar experiences intermittent network connectivity and performance issues across campus facilities, including hostel Wi-Fi zones, department Ethernet connections, routers, switches, and lecture halls. Such issues can arise from high latency, packet loss, jitter, insufficient bandwidth, excessive link utilization, device CPU overload, egress queue drops, interface errors, weak wireless signals, or hardware failures.

Diagnosing these issues is challenging because relevant information is distributed across different network devices and monitoring systems. Network telemetry may be incomplete, inconsistent, outdated, noisy, or represented using different data formats and terminology. Identifying the underlying cause often requires correlating evidence from multiple sources.

The challenge is to develop an interoperable multi-agent AI system that uses specialized agents to analyze different aspects of network health, exchange information, coordinate findings, and collaboratively identify the probable root cause.

2. What to Build

Build a multi-agent system in which specialized agents analyze different network diagnostic responsibilities. Agents must exchange structured information, interpret one another's findings, and combine evidence into a unified diagnostic report. Teams may use rule-based logic, existing AI models, LLMs, or hybrid approaches.

- Identify the probable root cause of a network issue.
- Identify affected devices or network segments where possible.
- Provide supporting evidence from telemetry and agent findings.
- Include alternative explanations or uncertainty where appropriate.
- Provide a confidence estimate or a clearly defined confidence category.
- Recommend practical corrective actions.

The system must not claim an exact root cause when the available evidence is insufficient.

3. Suggested Agent Architecture

- **Telemetry and Performance Agent:** Analyzes latency, jitter, packet loss, throughput, bandwidth, and link utilization.
- **Infrastructure Health Agent:** Examines device status, CPU and memory utilization, interface errors, queue drops, and other infrastructure metrics.
- **Connectivity Analysis Agent:** Examines topology, affected devices, link relationships, and the scope of connectivity failures.
- **Fault Correlation Agent:** Combines findings, compares competing explanations, and identifies the most likely root cause.
- **Recommendation and Reporting Agent:** Produces the final report, supporting evidence, uncertainty, and recommended corrective actions.

These roles are examples, not rigid requirements. At least two functional agents with distinct responsibilities must collaborate in the minimum working prototype. Four to five agents are encouraged

4. Mandatory Components & Optional Enhancements

Component	Status	Expected implementation
Multiple collaborating agents	**MANDATORY**	At least two functional agents exchange information and contribute to a shared diagnosis; four to five are encouraged.
Agent interoperability	**MANDATORY**	Document message formats, field meanings and units; provide a way to exchange and interpret findings.
Agent orchestration	**MANDATORY**	Coordinate agent tasks and consolidate findings into one result.
Network fault diagnosis	**MANDATORY**	Analyze supplied telemetry and identify probable root causes with supporting evidence.
Unified diagnostic report	**MANDATORY**	Include probable cause, evidence, affected devices where identifiable, uncertainty and recommended actions.
Basic validation and error handling	**MANDATORY**	Handle relevant malformed, missing or incompatible inputs and report failures clearly.
Reproducible testing and documentation	**MANDATORY**	Provide sample inputs, test scenarios, setup instructions and a demonstration.
Model Context Protocol (MCP)	**OPTIONAL**	May be used to expose diagnostic tools, telemetry or contextual resources to agents.
Human-in-the-Loop (HITL) workflow	**OPTIONAL**	No dedicated approval workflow is required; the core system produces recommendations for human review.
Persistent workflow state / pause-resume	**OPTIONAL**	Durable state, recovery and resumable sessions may be implemented. Basic message and finding logs are encouraged.
Interactive dashboard	**OPTIONAL, ENCOURAGED**	A dashboard may visualize telemetry, agent messages and results. A CLI is acceptable.
Automated remediation	**OPTIONAL,**	May simulate corrective actions. Do not modify production campus infrastructure without explicit authorization.

Important: *MCP, HITL, and durable workflow persistence are not prerequisites for a successful submission. Teams may use a custom JSON-based protocol, REST APIs, framework-provided messaging, or another documented approach. The central focus is reliable multi-agent interoperability and evidence-based diagnosis.*

5. Data and Testing Environment

Participants will use synthetic network telemetry, organizer-provided scenarios, or data collected in an authorized test environment. Access to the production campus network is not required.

Sample- Input-generator-code-COLAB [Utilize this code to generate synthetic network telemetry]

The dataset may include:

- Latency, jitter, packet loss, throughput, bandwidth, and link utilization.
- Device ID and device status.
- CPU and memory utilization, egress queue drops, and interface errors.
- Wireless signal measurements where available.
- Timestamps and network topology information.

Suggested test scenarios:

- **Normal network operation:** Measurements remain within expected ranges.
- **Link congestion:** High utilization, elevated queue drops, and increased latency.
- **Weak Wi-Fi signal:** Poor signal quality and increased retransmissions.
- **Ethernet or interface fault:** Rising interface errors, link instability, or packet loss.
- **DNS or DHCP failure:** Service-specific connectivity problems.
- **Interoperability failure:** Incompatible field names, units, schema versions, or missing message information.

6. Expected Output Format

The application must generate a structured JSON report. A human-readable Markdown report or dashboard may be provided in addition. The following is an illustrative example:


```
{
  "incident_id": "INC-8832",
  "status": "diagnosed",
  "suspected_root_cause": "Likely link congestion causing egress queue drops",
  "confidence": 0.86,
  "affected_devices": ["HOSTEL_CORE_SW_01"],
  "supporting_evidence": [
    {
      "metric": "link_utilization_pct",
      "value": 94,
      "interpretation": "High link utilization"
    },
    {
      "metric": "queue_drops",
      "value": 160,
      "interpretation": "Elevated packet drops"
    },
    {
      "metric": "latency_ms",
      "value": 280,
      "interpretation": "Elevated network latency"
    }
  ],
  "alternative_causes": ["Upstream link congestion"],
  "recommended_actions": [
    "Inspect upstream and egress link utilization",
    "Review queue configuration and traffic distribution"
  ],
  "evidence_sources": ["performance_agent", "infrastructure_agent"],
  "limitations": ["Root cause requires further verification"]
}
```


The values are illustrative and do not represent actual campus network measurements. Confidence should be based on a documented scoring method or calibrated model. Teams should be able to report insufficient evidence when a reliable diagnosis cannot be made.

7. Deliverables

1. **Source code repository:** A GitHub repository containing implementation, dependencies, setup instructions, and test scripts.
2. **Architecture documentation:** A README describing agent responsibilities, communication interfaces, orchestration, evidence correlation, error handling, and workflow.
3. **Sample data and test cases:** Inputs demonstrating normal operation and multiple network fault scenarios.
4. **Demo video:** A three-to-five-minute recording demonstrating an end-to-end diagnosis, inter-agent communication, and the final report.
5. **Working prototype:** A runnable application with a CLI or optional dashboard demonstrating the complete workflow.

8. Evaluation Criteria

Criterion	Points
Agent interoperability and protocol design	**30%**
Diagnostic accuracy and evidence quality	**25%**
Agent orchestration and conflict resolution	**15%**
Robustness to missing, inconsistent, or incompatible data	**15%**
Code quality, documentation, and demonstration	**5%**
Innovation, usability, and extensibility	**10%**
TOTAL	**100%**

9. Minimum Acceptance Criteria

- At least two collaborating agents with distinct responsibilities.
- Structured communication between agents.
- Evidence from multiple agents contributes to a single diagnosis.
- At least one malformed, missing, or incompatible input case is handled.
- A report containing probable root cause, supporting evidence, and recommended action.
- A reproducible run using supplied or documented test data.

10. Expected Impact

NetCoop AI aims to demonstrate how independently functioning AI agents can collaborate to simplify network fault diagnosis. By combining specialized analysis, interoperable communication, and evidence-based reasoning, the system should help network operators understand likely causes of connectivity problems more efficiently and make better-informed troubleshooting decisions.

The primary innovation is not simply using AI to detect a network fault. It is demonstrating how multiple specialized agents can exchange information, reconcile their findings, and produce one coherent and explainable diagnostic result.